

# Indian Institute of Technology (IIT-Bombay)

AUTUMN Semester, 2025

COMPUTER SCIENCE AND ENGINEERING

CS230: Digital Logic Design and Computer Architecture

Tutorial - II

Full Marks: 0

Time allowed:  $\infty$  hours

1. The output of a combinational circuit is one when all of its inputs are zero. What could be that gate? (more than one possibilities)

2. A gate T has logical properties that are defined by the map:

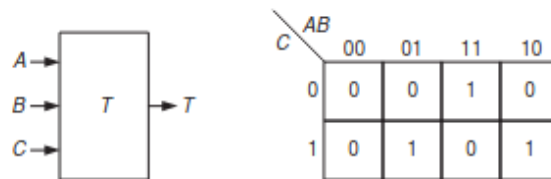


Figure 1: T-Gate

Prove that if the logic value 1 is given then any switching function can be realized by means of T gates, that is, T gates plus the logic value 1 are functionally complete.

3. Implement this following 4 variable Boolean Function  $F(A,B,C,D)$  with the help of a 8x1 Multiplexer and an OR GATE.

$$F = (C \cdot \bar{B}) + (A \cdot D \cdot \bar{B}) + (C \cdot \bar{A} \cdot \bar{D}) + (A \cdot B \cdot \bar{C} \cdot \bar{D}) + (B \cdot D \cdot \bar{A} \cdot \bar{C})$$

4. Express the following 4-variable Boolean Function

$$F = \Sigma(0, 2, 5, 7, 8, 10, 13, 15)$$

using:

- The minimum number of NOR gates
- The minimum number of NAND gates

5. An SR flip-flop, constructed from two cross-coupled NOR gates, is initially in the invalid state where both  $S=1$  and  $R=1$ . The inputs  $S$  and  $R$  are then simultaneously switched to 0 (the "hold" state). What is the final, stable state of the output  $Q$ ?

6. Figure 2 shows a sequential circuit. Analyze the circuit and answer the following questions

- Write down the excitation and output functions.
- Derive the excitation and state tables.

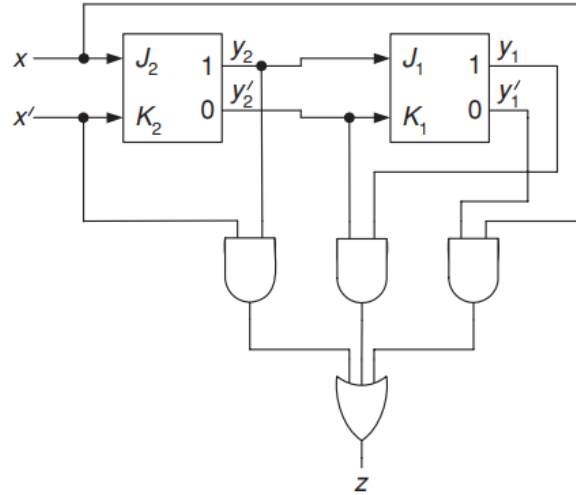


Figure 2

7. A long input sequence enters a one-input one-output synchronous sequential circuit, that is required to produce an output symbol  $z = 1$  whenever the sequence 1111 occurs. Overlapping sequences are accepted; for example, if the input sequence is 01011111 ..., the required output sequence is 00000011 ...

- Draw a state diagram.
- Select an assignment and show the excitation and output tables.

8. In the sequential circuit shown below in figure 3, if the initial value of the output  $Q_1Q_0$  is 00, what are the next four values of  $Q_1Q_0$ ?

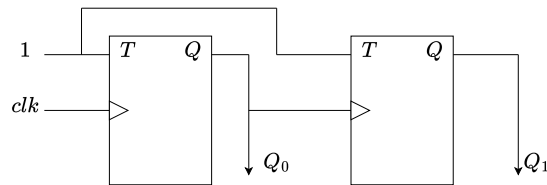


Figure 3

9. Consider the synchronous sequential circuit made of two T flip-flops and one D flip-flop shown below. All flip-flops share the same clock input  $\text{CLKIN}$ . Initially, the flip-flop outputs are  $(Q_1, Q_2, Q_3) = (0, 1, 1)$ .

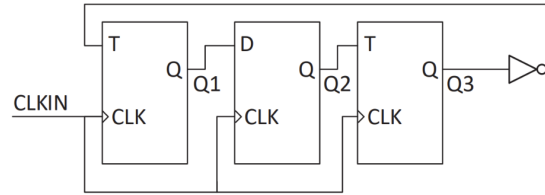
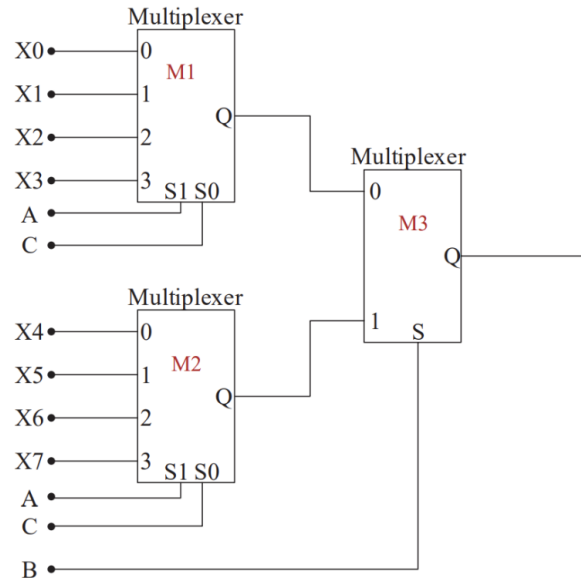


Figure:  $Q_1$  is the output of the left T-FF,  $Q_2$  of the D-FF, and  $Q_3$  of the right T-FF. Connections: the D input of the D-FF is  $Q_1$ ; the T input of the right T-FF is  $Q_2$ ; the T input of the left T-FF is  $\overline{Q_3}$ .

Which of the following values of  $(Q_1, Q_2, Q_3)$  can *never* be obtained at the outputs of this circuit?

- A.  $(1, 0, 0)$
- B.  $(0, 0, 1)$
- C.  $(1, 0, 1)$
- D.  $(1, 1, 1)$

18. A Boolean digital circuit is composed using two 4-input multiplexers ( $M_1$  and  $M_2$ ) and one 2-input multiplexer ( $M_3$ ) as shown in the figure.  $X_0, \dots, X_7$  are the inputs of the multiplexers  $M_1$  and  $M_2$ , and could be connected to either 0 or 1. The select lines of the multiplexers are connected to Boolean variables  $A, B, C$  as shown.



Circuit with MUXes  $M_1, M_2, M_3$ .

Which one of the following set of values of  $(X_0, X_1, X_2, X_3, X_4, X_5, X_6, X_7)$  will satisfy the Boolean function

$$f = \overline{A} + \overline{A}\overline{C} + A\overline{B}C?$$

- A. (1, 1, 0, 0, 1, 1, 1, 0)
- B. (1, 1, 0, 0, 1, 1, 0, 1)
- C. (1, 1, 0, 1, 1, 1, 0, 0)
- D. (0, 0, 1, 1, 0, 1, 1, 1)

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